

23CS412	EMBEDDED SYSTEMS AND IOT	L	T	P	C
		3	0	2	4
Category	Professional Core				
Prerequisites	Nil				
Course Objectives					
The course is intended to - Understand the fundamentals of embedded design architectures and challenges. - Explore the peripheral interfacing with embedded systems - Explore various design tools solutions for various programming phases. - Understand IoT system design - Understand IoT applications.					
Course Outcomes(COs)					
On successful completion of the course, students will be able to					
CO. No.	Course Outcome				Bloom's Level
CO 1	Explain fundamental embedded systems design paradigms, architectures, possibilities, and challenges, both with respect to software and hardware.				Understand
CO 2	Describe the hardware architecture and features of embedded microcontrollers and peripherals				Apply
CO 3	Explain software design tools and embedded system design programming phases.				Apply
CO 4	Describe IoT Architectures and Build simple IoT Systems using embedded target boards				Understand
CO5	Exhibit understanding of IoT applications.				Understand
Course Contents					
Unit – I	Embedded System Design				9
Introduction to embedded processors- Application Areas- Categories of embedded processors- Challenges in Embedded System Design, Real-time operating systems: OS Services, Process management, Timer functions, Event functions, Memory management, Device, file and IO subsystems management, Interrupt routines in RTOS environment and handling of interrupt source calls, Real-time operating systems, Basic design using an RTOS, RTOS task scheduling models, interrupt latency and response of the tasks as performance metrics, OS security issues. Introduction to embedded software development process and tools, Host and target machines, Linking and location software.					
Unit – II	Peripheral Interfacing				9
I/O Devices-Timers and Counters- Watchdog Timers, Interrupt Controllers- A/D and D/A, Interfacing Memory interfacing with a case study- I/O Device Interfacing with case Study- Programmed IO-Memory Mapped IO, Interfacing Protocols-SPI, I2C, USB, CAN.					
Unit – III	Embedded System Programming				9
Application Software, System Software, Design techniques – State diagrams, sequence diagrams, flowcharts, etc., Model-based system engineering (MBSE), Use of High-Level Languages embedded C / C++ Programming, Integrated Development Environment tools- Editor- Compiler- Linker- Automatic					

Code Generators- Debugger- Board Support Library- Chip Support Library, Analysis and Optimization Execution Time- Energy & Power.		
Unit – IV	IoT Design	9
Definition and characteristics of IoT, Technical Building blocks of IoT, Communication Technologies, Physical design of IoT - system building blocks - sensors - types of sensors - sensor Node and interfacing using Raspberry Pi / Arduino, Benefits and impact of IoT in Cybersecurity		
Unit – V	IoT Applications	9
IoT applications in home, infrastructures, buildings, security, Industries, Home appliances, other IoT electronic equipment, Industry 4.0 concepts.		
45 Periods		
Practical Exercises		
Ex. No.	Practical Exercises	Bloom's Level
1	Explore microcontrollers based programs using Embedded C.	Understand
2	Write Basic and arithmetic Programs Using Embedded C.	Apply
3	Write Embedded C program to test interrupt and timers.	Apply
4	Develop Real time applications – clock generation, waveform generation, counter using embedded C.	Apply
5	Explore different communication methods with IoT devices.	Apply
6	To interface LED/Buzzer with platform/ Arduino /Raspberry Pi. and write an embedded C program to turn on / off LED/Buzzer with specified delay.	Apply
7	To interface DC/stepper motor using relay with open platform/ Arduino /Raspberry Pi and write an embedded C program to turn on the motor if push button is pressed.	Apply
8	Develop simple IoT Applications using Raspberry Pi / Arduino	Apply
9	Develop a program to display temperature and humidity readings in LCD.	Apply
10	Develop IoT Applications using Arduino / Raspberry Pi.with sensors	Apply
30 Periods		
Total : 75 Periods		
Textbooks		
1. James K Peckol, Embedded Systems – A Contemporary Design Tool, John Wiley, 2008. 2. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017. 3. Venkata Krishna, Sasikumar Gurumoorthy, Mohammad S. Obaidat, Internet of Things and Personalized Healthcare Systems”, Springer Briefs in Applied Sciences, and Technology, Forensic and Medical Bioinformatics, 2019.		
Reference Books		
1. Shibu K V, Introduction to Embedded Systems, Tata McGraw Hill Education Private Limited, 2009. 2. David E.Simon, Embedded Software Primer, Addison Wesley, ISBN-13: 978-0201615692. 3. Barry B.Brey, The Intel Microprocessors, Architecture, Programming and Interfacing” Barry B.Brey, 6th Edition, Pearson Education.		
e-Resources:		

1. <https://archive.nptel.ac.in/courses/128/108/128108016/>, “IoT and Embedded Systems”, Prof. Deberata Das, IIT Bengaluru.
2. <https://archive.nptel.ac.in/courses/108/105/108105057/>, “Embedded Systems”, Prof. Rajib Mall, IIT Kharagpur.

Mapping of Course Outcomes (COs) with Programme Outcomes (POs) Programme Specific Outcomes (PSOs)															
COs	POs												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO 1	2	3	1	-	-	-	-	-	-	1	-	1	-	-	-
CO 2	2	3	1	-	-	-	-	-	-	1	-	1	-	-	-
CO 3	2	3	1	-	-	-	-	-	-	1	-	1	-	-	-
CO 4	2	3	1	-	-	-	-	-	-	1	-	1	-	-	-
CO 5	2	3	1	-	-	-	-	-	-	1	-	1	1	-	-
Average	2	3	1	-	-	-	-	-	-	1	-	1	1	-	-

3 – High 2 – Medium 1- Low '-' - Correlation

Assessment Components	Duration	Syllabus to be Covered	Max. Marks	Weightage for Internal Marks	Continuous Internal Assessment Marks	End Semester Examination Marks
CIA I	3 hours	2.5 units	100	10	20	50
CIA II	3 hours	2.5 units	100	10		
Practical Component						
Observation & Analysis of Experimental results, Viva Voce, Quiz based on rubrics		All Experiments	75	22.5	30	-
Model Exam	3 Hours		25	7.5		
Total					50	50